



MRS6240-AIP-DEV-V1

mmWave Radar Development Board

User Guide

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1 Overview

The RS6240-AIP-DEV-BOX development kit is an application development platform launched by Possumic Technology, integrating ultra-low power 60GHz FMCW millimeter-wave radar, Bluetooth 5 (LE), and IEEE 802.15.4 wireless functions into an intelligent radar sensing chip. It is designed to facilitate developers in scheme evaluation and design. This development kit includes the MRS6240 series module (MRS6240-P2512 module) and an MRS6240-AIP-DEV-V1 development board base, allowing developers to connect a wide range of peripherals to meet the development needs of various application scenarios. It is particularly suitable for applications such as smart home, smart appliances, smart security, and smart lighting. Its multiple Rx/Tx antenna feature brings excellent detection experience to product applications.

This document primarily introduces the usage of the MRS6240-AIP-DEV-V1 development board baseboard, encompassing component information, power supply information, interface information, and other related introductions of this evaluation baseboard.

2 Hardware Introduction

2.1 Hardware block diagram

The main components and application connection methods of MRS6240-AIP-DEV-V1 are shown in the figure below:

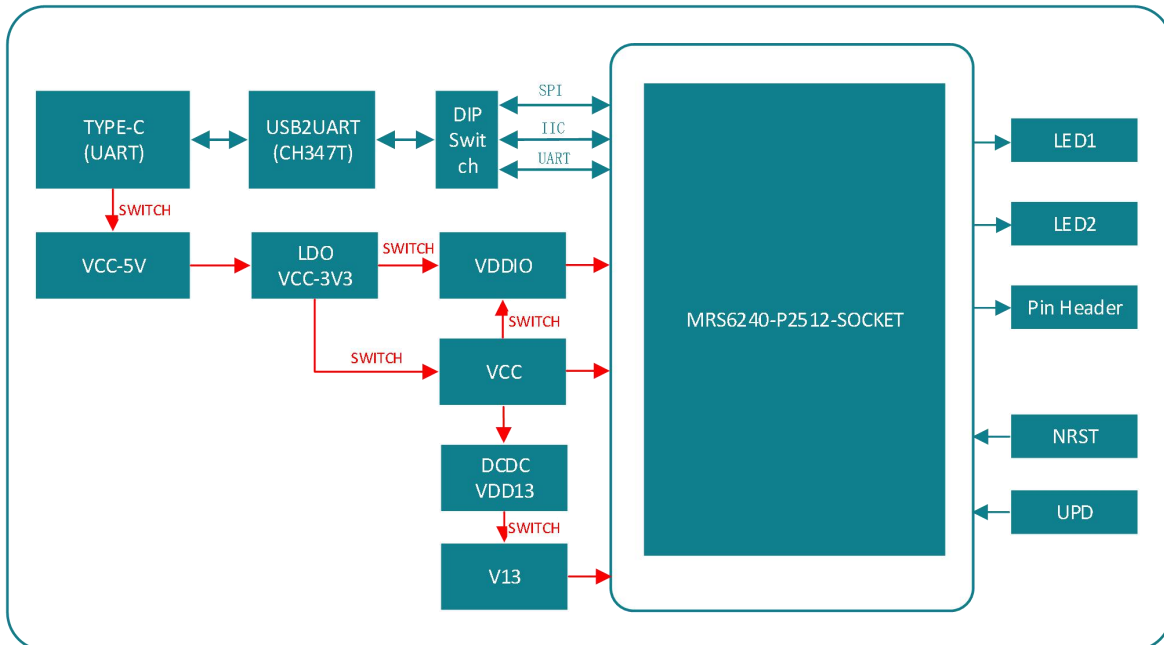


Figure 2-1 Schematic Block Diagram of MRS6240-AIP-DEV-V1 Hardware Connection

2.2 Component introduction

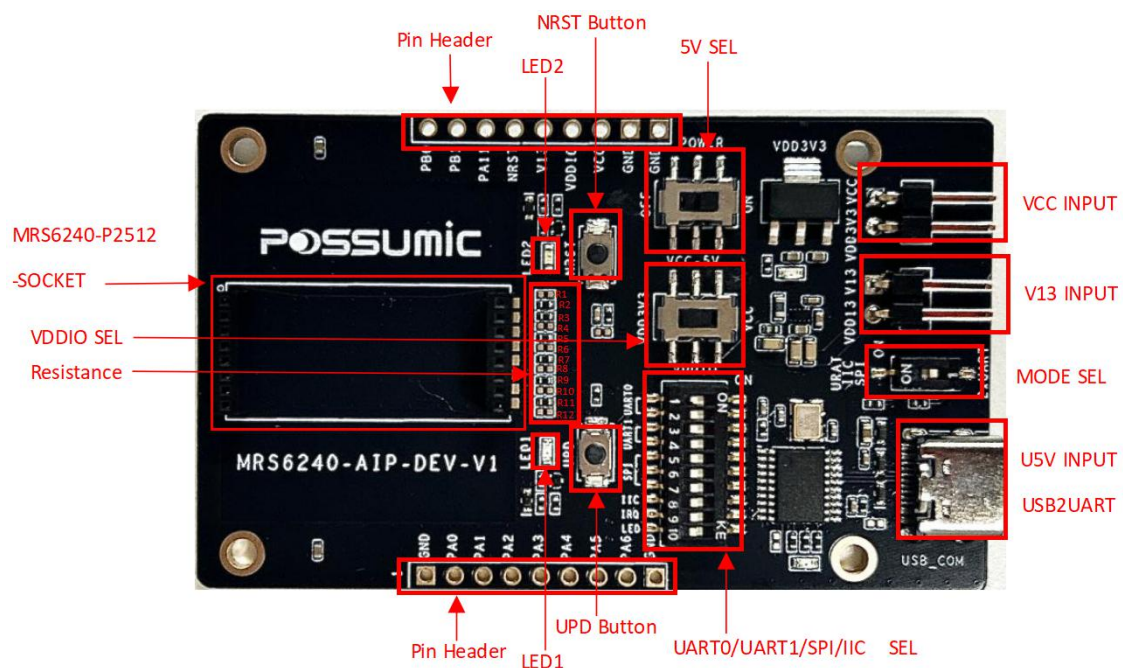


Figure 2-2 Physical Image of Development Board Components

Table 2-1 Description of Development Board Components

Component		Instructions
Module Interface	MRS6240-P2512-SOCKET	Used for connecting the millimeter-wave radar module MRS6240-P2512
Button	UPD Button	Upgrade download button In the power-off state, first press the UPD Button, then plug in the USB cable to power on, the chip will start and enter "Firmware Download" mode, and then download the firmware through the serial port UART0.
	NRST Button	Module reset button
Power Supply	U5V INPUT	Type-C VBUS 5V power supply, and also serves as a UART to USB conversion path, connected to the USB port of a PC.
	5V SEL	Switch for the main power supply VCC-5V of the development board: ➤ Turn to "ON" (default): Turn on the VCC-5V power supply. ➤ Turn to "OFF": Turn off the VCC-5V power supply.
	VDDIO SEL	Selection of the source for module power supply VDDIO, which is used to power the module IO and peripherals: ➤ Set to "VCC" (default): Use VCC as the power supply ➤ Set to "VDD3V3": Use VDD3V3 as the power supply
	VCC INPUT	Module VCC power supply switch, VCC is the main power supply for the radar module: ➤ Connect the jumper cap (default): Use VDD3V3 supplied by the LDO output. The LDO is a component on the development board, and its input power supply is VCC-5V. ➤ Remove the jumper cap: If an external DC power supply is required, remove the jumper cap and connect the external power supply to the VCC pin of the Pin Header.
	V13 INPUT	Power supply selection for module V13, where V13 serves as the main power source for the RF part of the radar module: ➤ Connect the jumper cap (default): Use VDD13 supplied by the DCDC output. The DCDC is a component on the development board, and its input power supply is VCC. ➤ Disconnect the jumper cap: Without VDD13 power supply, the radar module will adaptively use the internal LDO, and the V13 network voltage will be 1.5V.
Other	USB2UART	Type-C USB interface, which is used for power supply and UART/IIC/SPI communication of the development board. When the MODE SEL is turned to OFF, the USB interface supports dual serial ports. After connecting to a PC, the device name is identified as "USB Serial Device COM*" (where * represents the serial port number, with the larger number corresponding to the radar's UART0 and the smaller number corresponding to the radar's UART1). When the MODE SEL is turned to ON, the USB interface supports single serial port/IIC/SPI. After connecting to a PC, the device name is recognized as "USB Serial Device COM*".

	LED	Apply indicator LED lights. The power supply comes from VDD3V3. The control signal for LED1 is PA6, and the control signal for LED2 is PA11.
	MODE SEL	CH347T working mode selection (dip switch): ➤ ON (default): USB to high-speed single serial port + USB to 2-wire IIC + USB to 4-wire SPI ➤ OFF: USB to high-speed dual serial ports Note: The direction of ON is subject to the mark on the dial switch
	UART0/UART1/SPI/IIC SEL	MRS6240 communication mode selection (dip switch): ➤ Set 1 and 2 to ON, and all others to OFF: UART0 is connected to Type-C ➤ Set pins 3 and 4 to ON, and all others to OFF: UART1 is connected to Type-C ➤ Set 5, 6, and 7 to ON, and all others to OFF: SPI is connected to Type-C ➤ 8 is set to ON, and the others are set to OFF: IIC is connected to Type-C ➤ 9 is set to ON, 10 is set to OFF: PA6 is used to report interrupt data to the USB chip ➤ 10 for ON, 9 for OFF: PA6 is used to light up LED1
	Pin Header	All available IO ports (excluding the SPI interface dedicated to Flash) and power supplies (VCC, VDDIO, V13) of the module are all brought out to this pin header.
	Resistance	If UART1 (PA0, PA1) needs to be used on the pin header, OR resistors need to be soldered on R1 and R4. If SPI (PA0, PA1, PA2, PA3) is required to be used on the pin header, it is necessary to solder OR resistors on R1, R4, R6, and R8. If IIC (PA4, PA5) is required to be used on the pin header, OR resistors need to be soldered on R10 and R12.

2.3 Power options

1 Development board power supply:

1) Power supply for USB_COM interface: Set the dip switch VCC-5V to ON (default);

2 There are two options for supplying power to the module VCC:

1) Development board VCC-3V3 power supply: VCC INPUT is short-circuited, meaning VDD3V3 and VCC (default);

2) External power supply: Power is supplied through the VCC pin of the Pin Header, with the VCC INPUT disconnected.

3 There are two options for VDDIO power supply of development boards and modules:

1) Development board VCC-3V3 power supply: Set the dip switch VDDIO to VDD-3V3 (default);

2) VCC power supply: Set the dip switch VDDIO to VCC.

4 There are two power supply options for module V13:

- 1) VINT power supply output by the module: V13 INPUT is disconnected (default);
- 2) Development board DCDC power supply: V13 INPUT short circuit.

2.4 Pin header

The names and functions of the pin headers on both sides of the development board are as follows:

Table 2-2 Definition and Description of Pin Header Signals

Serial Number	Name	Type	Function
1	GND	G	Ground Pin, Short directly to board ground plane
2	PA0	IO/WUP	PA0, WIO0, SPI0_CS, UART1_TX, CANFD_TX, PWM0
3	PA1	IO/WUP	PA1, WIO1, SPI0_CLK, UART1_RX, CANFD_RX, PWM1
4	PA2	IO/WUP	PA2, WIO2, SPI0_MOSI, UART1_RTS, I2C0_SDA, UART2_TX/LIN_TX, PWM2
5	PA3	IO/WUP	PA3, WIO3, SPI0_MISO, UART1_CTS, I2C0_SCL, UART2_RX/LIN_RX, PWM3
6	PA4	IO/WUP	PA4, WIO4, SPI0_HOLD, I2C0_SDA, PWM4
7	PA5	IO/WUP	PA5, WIO5, SPI0_WP, I2C0_SCL, PWM5
8	PA6	IO/WUP	PA6, WIO6, UART1_RTS, UART2_TX, I2C1_SDA, PWM6, GPADC6
9	GND	G	Ground Pin, Short directly to board ground plane
10	GND	G	Ground Pin, Short directly to board ground plane.
11	GND	G	Ground Pin, Short directly to board ground plane.
12	VCC	PI	Development board VCC power supply
13	VDDIO	PI	Development board VDDIO power supply
14	V13	PI/PO	The development board V13 power supply provides 1.3V when using an external DCDC, and 1.5V when using an internal LDO.
15	NRST	I	Radar module reset signal
16	PA11	IO/WUP	PA11, WIO11, I2C1_SCL, CANFD_RX, PWM3, GPADC3
17	PB1	IO/WUP	PB1, WIO17, UART0_RX, PWM1
18	PB0	IO/WUP	PB0, WIO16, UART0_TX, PWM0

- **G** Represents the ground
- **PI** Stands for Input Power
- **PO** Stands for output power
- **GPIO** Stands for General-Purpose Digital Input/Output
- **I** Represents the input pin
- **WUP** Stands for a digital pin that can maintain a state or wake up the system in standby mode, where WIOx refers to the sequential number of the WUP type IO.

3 Hardware Usage

3.1 Required hardware

Hardware preparation for using the MRS6240-AIP-DEV-V1 development board:

- Radar module MRS6240-P2512;
- Type-C USB data cable;
- Computer, and install the CH347T serial port driver;

3.2 Module connection

The radar module includes MRS6240-P2512. The connection method is shown in the figure below.

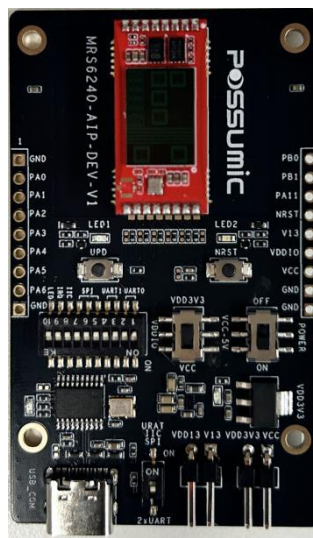


Figure 3-1 P2512

3.3 Basic usage method

- **Step 1:** Configure the development board power supply according to the default connection method described in [the power supply section of Table 2-1 Description of Development Board Components instructions](#).
- **Step 2:** Configure the signal path according to the default connection methods described in [Table 2-1, specifically for MODE SEL and UART0/UART1/SPI/IIC SEL](#) in the development board component instructions.
- **Step 3:** Insert the radar module MRS6240-P2512, paying attention to the orientation of PIN1 of the module. Both the module and the development board have corresponding white dot markings.
- **Step 4:** Connect the PC to the U5V TYPE-C port of the development board via a Type-C data cable. Two serial port devices will appear on the computer side, with the larger serial port number corresponding to the UART0 of the radar and the smaller serial port number corresponding to the UART1 of the radar.
- **Step 5:** Operate using the PC host computer tool.

3.4 Power consumption measurement method

- **Step 1:** Set "5V SEL" to "ON", set "VDDIO SEL" to "VCC", and disconnect the jumper cap of "VCC INPUT".

- **Step 2:** If you need to measure the operating condition with DCDC, connect the jumper cap of "V13 INPUT"; if you need to measure the operating condition without DCDC, disconnect the jumper cap of "V13 INPUT".
- **Step 3:** Use a TYPE-C data cable to supply power to the U5V TYPE-C port.
- **Step 4:** Use an external high-precision power measurement instrument to supply power to VCC and GND of the PIN Header, and record the current data.



Figure 3-5 Power measurement instrument and hardware configuration

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